Chebotarev density theorem (complete author Blueprint proof chain)

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Editorial note. Author Blueprint proof, not a Lean formalization. The native source at this pinned commit contains this proof entirely in blueprint_comment blocks. Source citations and any author omissions are retained without mathematical repair.

Allowed standard prerequisites (editorial).

- Author citations to Sharifi Sections 7.1-7.2, as retained in the original proof.

1 The Chebotarev density theorem: the case of cyclotomic extensions

Throughout this section, K is a number field, $m \geq 1$ is a fixed integer, $L = K(\mu_m)$, and $G = \mathrm{Gal}(L/K)$. (In particular G is abelian, and for $K = \mathbb{Q}$ one recovers $G \cong (\mathbb{Z}/m\mathbb{Z})^\times$.) Write ζ_K and ζ_L for the Dedekind zeta functions of K and L, and for an abelian character $\chi : G \to \mathbb{C}^\times$ write $L(\chi, s)$ for the associated Artin L-function (equivalently, the Hecke L-function of the ideal character attached to χ; cf. Notation 7.1.17 and Proposition 7.1.18 of `https://www.math.ucla.edu/~sharifi/algnum.pdf`).

The goal of this section is to prove the Chebotarev density theorem in the cyclotomic case (Proposition 1.1 below), following the classical argument via Artin L-functions as in Sharifi, Propositions 7.1.16–7.1.19 and Proposition 7.2.1. The abelian and general cases are then reduced to this one in the subsequent sections.

Lemma 1.1 (Artin L-function as Euler product)**.** Let $\chi : G \to \mathbb{C}^\times$ be an abelian character. For $\Re(s) > 1$ one has

$$L(\chi, s) \;=\; \prod_{\mathfrak{p}} \bigl(1 - \chi(\mathfrak{p})\, N\mathfrak{p}^{-s}\bigr)^{-1} \;=\; \sum_{\mathfrak{a} \subseteq \mathcal{O}_K} \chi(\mathfrak{a})\, N\mathfrak{a}^{-s},$$

where the product runs over nonzero prime ideals of $\mathcal{O}_K$, with the convention $\chi(\mathfrak{p}) = 0$ if $\mathfrak{p}$ ramifies in the fixed field of $\ker \chi$.

Proof. This is the specialisation of the Artin Euler product (Definition 7.1.15 of Sharifi) to abelian characters; see Proposition 7.1.18 of `https://www.math.ucla.edu/~sharifi/algnum.pdf`. Absolute convergence for $\Re(s) > 1$ follows from comparison with $\zeta_K(s)$. □

Lemma 1.2 (Dedekind-factor)**.** For $\Re(s) > 1$ one has

$$\zeta_L(s) \;=\; \prod_{\chi : G \to \mathbb{C}^\times} L(\chi, s),$$

where the product runs over all (necessarily one-dimensional) irreducible characters of G.

Proof. In general, for a Galois extension L/K and characters χ of irreducible representations of $\mathrm{Gal}(L/K)$, Proposition 7.1.16 of `https://www.math.ucla.edu/~sharifi/algnum.pdf` gives $\zeta_L(s) = \prod_\chi L(\chi, s)^{\chi(1)}$ for $\Re(s) > 1$. In the present abelian (cyclotomic) setting every irreducible character is one-dimensional, so $\chi(1) = 1$ and the exponents disappear. Alternatively, comparing Euler factors at an unramified prime $\mathfrak{p}$: the primes of L above $\mathfrak{p}$ contribute the factor $(1 - N\mathfrak{p}^{-fs})^{-g}$ to ζ_L, while the Artin factors multiply to the same quantity by the usual identity $\prod_\chi (1 - \chi(\varphi_\mathfrak{p})X) = 1 - X^f$ (with $X = N\mathfrak{p}^{-s}$ and f the residue degree). □

Lemma 1.3 (Simple pole)**.** The Dedekind zeta function ζ_L admits a meromorphic continuation to a neighbourhood of the line $\Re(s) = 1$ with a single simple pole at $s = 1$ (and is otherwise holomorphic and nonvanishing on that line after removing the pole). In particular $\log \zeta_L(s) \sim \log(s-1)^{-1}$ as $s \to 1^+$.

Proof. This is the standard analytic continuation of Dedekind zeta (Theorem 7.1.12 of `https://www.math.ucla.edu/~sharifi/algnum.pdf`): absolute convergence of the ideal Dirichlet series and Euler product for $\Re(s) > 1$, meromorphic continuation across $\Re(s) > 1 - [L : \mathbb{Q}]^{-1}$, and a simple pole at $s = 1$. The logarithmic asymptotic is immediate from the simple pole. □

Lemma 1.4 (Continuation of nontrivial Artin L-functions)**.** Let $\chi : G \to \mathbb{C}^\times$ be a nontrivial character. Then $L(\chi, s)$ extends to an analytic function on the half-plane $\Re(s) > 1 - [K : \mathbb{Q}]^{-1}$, and in particular is holomorphic at $s = 1$.

Proof. Following Proposition 7.1.19 of `https://www.math.ucla.edu/~sharifi/algnum.pdf`: let n be the order of χ. Geometry of numbers supplies the estimate that, for each nth root of unity ζ, the number of ideals $\mathfrak{a}$ with $N\mathfrak{a} \le N$ and $\chi(\mathfrak{a}) = \zeta$ is $CN + O\big(N^{1-[K:\mathbb{Q}]^{-1}}\big)$ with C independent of ζ. Summing against ζ cancels the main terms, so $\sum_{N\mathfrak{a} \le N} \chi(\mathfrak{a}) = O\big(N^{1-[K:\mathbb{Q}]^{-1}}\big)$. Lemma 7.1.5 of Sharifi then yields absolute uniform convergence of $\sum_\mathfrak{a} \chi(\mathfrak{a})\, N\mathfrak{a}^{-s}$ on compact subsets of $\Re(s) > 1 - [K : \mathbb{Q}]^{-1}$. □

Lemma 1.5 (Dedekind-nonvanishing)**.** For any nontrivial character χ of $G = \mathrm{Gal}(L/K)$, one has $L(\chi, 1) \neq 0$. Moreover $L(\chi, s)$ does not vanish for $\Re(s) = 1$.

Proof. For the value at $s = 1$: write $\log \zeta_L(t) = \sum_\chi \log L(\chi, t)$ for real $t > 1$. As $t \to 1^+$, the left side is $\sim \log(t-1)^{-1}$ by Lemma 1.3. If some nontrivial $L(\chi, \cdot)$ had a zero of order $m_\chi \ge 1$ at $s = 1$, the right side would behave like

$\big(1-\sum_\chi m_\chi\big)\log(t-1)^{-1}$ up to a bounded error (the trivial character contributes the pole of ζ_K, absorbed into the factorisation), forcing $1-\sum m_\chi \le 0$, a contradiction. Hence $L(\chi,1)\neq 0$ for all nontrivial χ (Sharifi, Proposition 7.1.19).

For the rest of the line $\Re(s)=1$, $s\neq 1$: adapt the classical nonvanishing argument for Dirichlet L-functions (comparison of $\zeta_L(s)$, $\zeta_L(s)^3\,|L(\chi,s)|^4$, or the $3+4\,\mathrm{Re}$ inequality) using the Euler product of Lemma 1.1 and the absence of poles of nontrivial $L(\chi,\cdot)$ on $\Re(s)=1$ from Lemma 1.4. □

Lemma 1.6 (Orthogonality for Frobenius classes)**.** Fix $\sigma\in G$. For every prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$ unramified in L,

$$\sum_{\chi:G\to\mathbb{C}^\times}\chi(\sigma)^{-1}\,\chi(\varphi_\mathfrak{p}) \;=\; \begin{cases}|G| & \text{if } \varphi_\mathfrak{p}=\sigma,\\ 0 & \text{otherwise.}\end{cases}$$

Equivalently, if $\sigma(\zeta_m)=\zeta_m^a$ with $\gcd(a,m)=1$, the sum equals $|G|$ precisely when $N\mathfrak{p}\equiv a \pmod m$.

Proof. Standard character orthogonality on the finite abelian group G. The reformulation in terms of $N\mathfrak{p}$ mod m uses that $\varphi_\mathfrak{p}(\zeta_m)=\zeta_m^{N\mathfrak{p}}$ for the cyclotomic character. □

Proposition 1.1 (Cyclotomic Chebotarev density)**.** For every $\sigma\in G$, the set of prime ideals $\mathfrak{p}$ of K unramified in L with Frobenius $\varphi_\mathfrak{p}=\sigma$ has Dirichlet density $1/|G|$.

Proof. This is Proposition 7.2.1 of `https://www.math.ucla.edu/~sharifi/algnum.pdf`. For $\Re(s)>1$ one has $\log L(\chi,s)\sim\sum_\mathfrak{p}\chi(\mathfrak{p})\,N\mathfrak{p}^{-s}$. Summing against $\chi(\sigma)^{-1}$ and applying Lemma 1.6 yields

$$\sum_\chi\chi(\sigma)^{-1}\log L(\chi,s)\;\sim\;|G|\sum_{\varphi_\mathfrak{p}=\sigma}N\mathfrak{p}^{-s}.$$

On the other hand, by Lemmas 1.2 and 1.5 the left side is $\sim\log\zeta_K(s)\sim\log(s-1)^{-1}$ as $s\to1^+$ (only the trivial character contributes a pole). Comparing the two asymptotics gives the asserted Dirichlet density. □

Lemma 1.7 (PNT for one character)**.** For any nontrivial character χ of G,

$$\sum_{N\mathfrak{p}\le x}\chi(\mathfrak{p})\,\log N\mathfrak{p}\;=\;o(x)\qquad(x\to\infty).$$

(Equivalently, writing Λ_χ for the von Mangoldt-type coefficients of $-\frac{L'}{L}(\chi,s)$, one has $\sum_{n\le x}\Lambda_\chi(n)=o(x)$.)

Remark on prime powers. Expanding $-\frac{L'}{L}(\chi,s)$ as a Dirichlet series selects coefficients supported on prime powers $\mathfrak{p}^j$ with weight $\chi(\varphi_\mathfrak{p})^j\log N\mathfrak{p}$. Thus an *exact* identity expressing $-\sum_\chi\chi(\sigma)^{-1}\frac{L'}{L}(\chi,s)$ as a sum over primes with Frobenius σ should use the condition $\varphi_\mathfrak{p}^j=\sigma$ rather than $\varphi_\mathfrak{p}=\sigma$. For the density statement in Proposition 1.1 this distinction is immaterial: the $j=1$ terms agree with the Frobenius condition, while the $j\ge2$ terms converge absolutely for $\Re(s)>1/2$ and do not affect the residue at $s=1$.

Proof. By Lemmas 1.4 and 1.5, $L(\chi, s)$ is holomorphic and nonvanishing on $\Re(s) \geq 1$, so $-\frac{L'}{L}(\chi, s)$ extends continuously to $\Re(s) \geq 1$. The claimed prime-sum estimate then follows by the same Wiener–Ikehara / Ingham contour argument used for Dirichlet L-functions in the prime-number theorem in arithmetic progressions (cf. the material already formalised for Dirichlet L-functions earlier in this file). Passing through Λ_χ (rather than the bare prime sum) makes the $j \geq 2$ prime-power contributions explicit and shows they are $O(x^{1/2} \log x)$, hence absorbable in the $o(x)$ error; cf. the remark after the statement. □

2 The Chebotarev density theorem: the case of abelian extensions

Now let L/K be an arbitrary finite abelian extension with Galois group G, and fix $\sigma \in G$. The goal is to show that the primes of K with Frobenius σ still have Dirichlet density $1/|G|$, by reducing to the cyclotomic case already treated. (Cf. Theorem 7.2.2, Step 2, of `https://www.math.ucla.edu/~sharifi/algnum.pdf`; cyclic extensions already suffice for the later reduction to the general case.)

Lemma 2.1 (Cyclotomic crossing)**.** Choose an integer $m \geq 1$ not dividing the discriminant of L/K, large enough that $H := \mathrm{Gal}(L(\mu_m)/L) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ via the cyclotomic character and $\mathrm{Gal}(L(\mu_m)/K) \cong G \times H$. For $\sigma \in G$ and $\tau \in H$ write $S_{\sigma,\tau}$ for the set of primes of K unramified in $L(\mu_m)$ with Frobenius $(\sigma, \tau) \in G \times H$, and S_σ for the set of primes of K unramified in L with Frobenius σ in G. Then

$$\delta_{\inf}(S_\sigma) \;=\; \sum_{\tau \in H} \delta_{\inf}(S_{\sigma,\tau}).$$

Proof. Every prime counted in S_σ lifts to primes in the various $S_{\sigma,\tau}$ according to the Frobenius in the cyclotomic layer; the identity of lower densities is the usual additivity of Dirichlet densities over a finite partition (up to the finitely many ramified primes). □

Lemma 2.2 (Density after cyclotomic crossing)**.** In the notation of Lemma 2.1, if the order of $\tau \in H$ is divisible by $|G|$, then $\langle(\sigma, \tau)\rangle \cap (G \times \{1\}) = \{1\}$, so $L(\mu_m)$ is obtained by adjoining μ_m to the fixed field $F := L(\mu_m)^{\langle(\sigma,\tau)\rangle}$. Applying Proposition 1.1 to the cyclotomic extension $F(\mu_m)/F$ and transporting densities as in the reduction step of Theorem 7.2.2 yields $\delta(S_{\sigma,\tau}) = 1/(|G|\,|H|)$.

Proof. See Sharifi, Theorem 7.2.2, Step 2 (display after "Now suppose that $|G|$ divides the order of τ"). □

Proposition 2.1 (Abelian Chebotarev)**.** Let L/K be finite abelian with Galois group G. For every $\sigma \in G$, the Dirichlet density of primes of K with Frobenius σ exists and equals $1/|G|$.

Proof. Let $H_n \subset H$ be the set of elements whose order is divisible by n. Summing the densities of Lemma 2.2 over $\tau \in H_n$ gives $\delta_{\inf}(S_\sigma) \geq |H_n|/(|G|\,|H|)$. Choosing m so that $|H_n|/|H|$ is arbitrarily close to 1 (possible since there are primes $m \equiv 1 \pmod{n^j}$ by the cyclotomic case already proved, and $|H_n|/|H|$ tends to 1 as $j \to \infty$), one obtains $\delta_{\inf}(S_\sigma) \geq 1/|G|$. Summing over $\sigma \in G$ forces equality throughout, so each density exists and equals $1/|G|$. □

3 The Chebotarev density theorem: the general case

Finally let L/K be an arbitrary finite Galois extension with group G, and let $C \subset G$ be a conjugacy class. The theorem reduces to the abelian (indeed cyclic) case already proved, by passing to the fixed field of an element of C.

Lemma 3.1 (Reduction to a cyclic subextension)**.** Fix $\sigma \in C$ and let E be the fixed field of $\langle\sigma\rangle$, so L/E is cyclic of degree $f = |\langle\sigma\rangle|$. Write S for the set of primes of K unramified in L whose Frobenius class equals C, and T_σ for the set of primes $\mathfrak{P}$ of E unramified in L (and lying over K) with Frobenius equal to σ. Then

$$\delta(S) \;=\; \frac{f\,|C|}{|G|}\,\delta(T_\sigma),$$

whenever either density exists. In particular, the Chebotarev statement for L/K and C follows from the abelian statement for the cyclic extension L/E and the element σ.

Proof. If $\mathfrak{P} \in T_\sigma$, then $\varphi_{\mathfrak{P}} = \sigma$ fixes E, so $\mathfrak{P}$ has residue degree one over K. There are exactly $|G|/f$ primes of L over $\mathfrak{P} \cap K$, and their Frobenii are equidistributed among the $|C|$ elements of the conjugacy class C; exactly $|G|/(f\,|C|)$ of them have Frobenius σ. Comparing the Dirichlet series $\sum N\mathfrak{p}^{-s}$ over S with $\sum N\mathfrak{P}^{-s}$ over T_σ (and using $\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \sim \sum_{\mathfrak{P}} N\mathfrak{P}^{-s}$) yields the displayed identity. See Sharifi, Theorem 7.2.2, Step 1. □

Theorem 3.1 (Chebotarev density theorem)**.** Let L/K be a finite Galois extension of number fields with Galois group G, and let $C \subset G$ be a conjugacy class. The set of prime ideals $\mathfrak{p}$ of K unramified in L whose Frobenius conjugacy class equals C has Dirichlet density $|C|/|G|$.

Proof. Combine Lemma 3.1 with Proposition 2.1 applied to the cyclic extension L/E: the latter gives $\delta(T_\sigma) = 1/f$, whence $\delta(S) = |C|/|G|$. □